When LLMs Evaluate LLMs in Hiring: A Multi-Model Auditing Framework for Generative Preference Bias in AI-Assisted Recruitment

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Motivation

Applicants write cover letters with AI. Employers screen with AI.

- Most job seekers now use LLMs to draft application material.
- 7 in 10 companies plan to deploy AI in screening.

OPEN QUESTION Once both sides use AI, does which model you use change your odds?

Two-Phase Audit Pipeline

PHASE 1

Generation

8 writer models draft cover letters (350–400 words, gender-neutral persona)
for 50 CVs × 10 jobs — stratified top-25 high-fit / next-25 moderate-fit by cosine similarity.

GPT-4o-mini | GPT-5-mini | Claude 4.5 Haiku | Gemini 3 Flash | Gemini 2.0 Flash | DeepSeek Chat | DeepSeek R1 8B | Llama 3.1 8B

API-hosted locally-hosted (open-weight)

PHASE 2

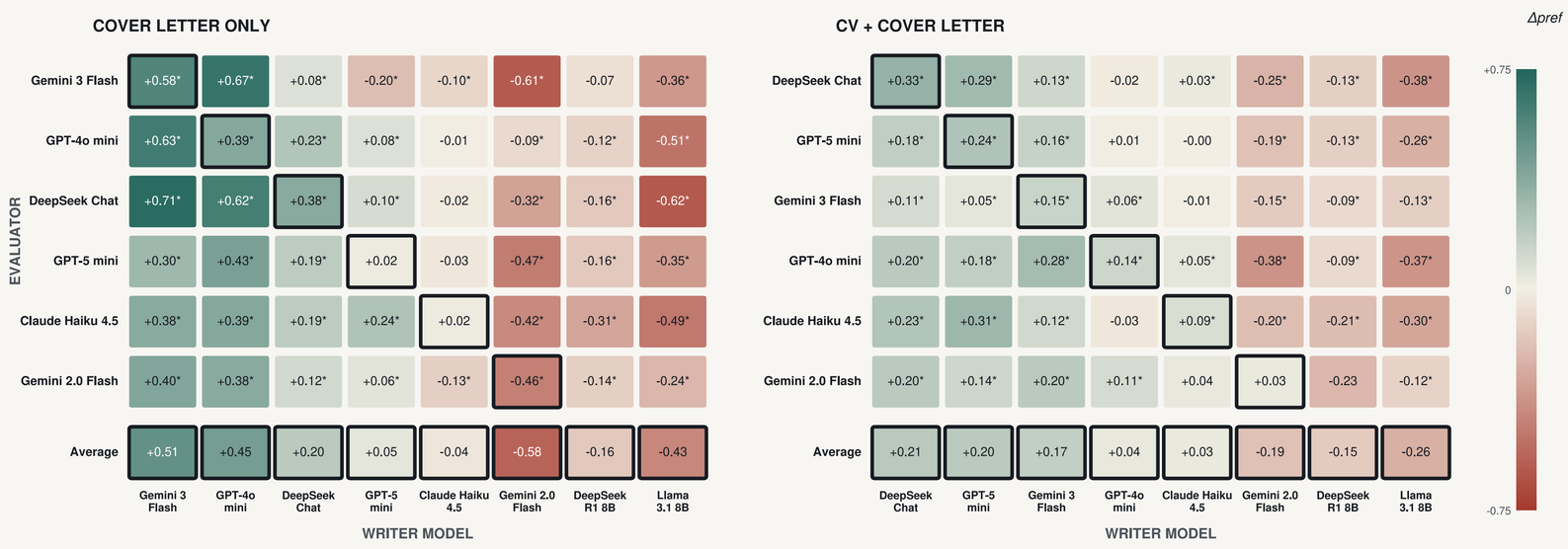
Evaluation

6 API-hosted evaluator models score every candidate, × 4 iterations each, under:

CV Only | Cover Letter Only | CV + Cover Letter

Results

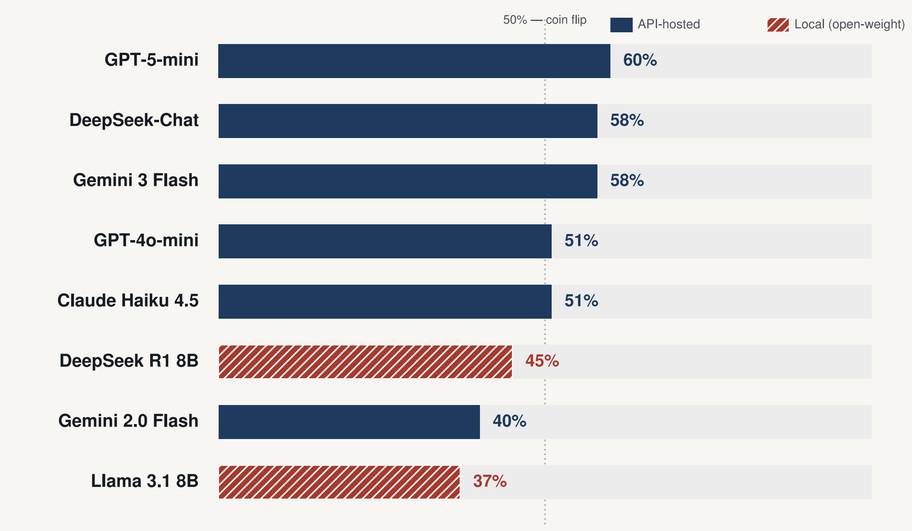
A Generative Preference Bias



B Evaluator Fairness

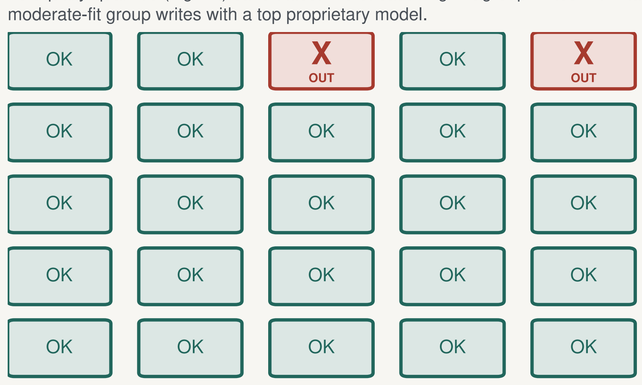


C Head-to-head Win Rate



D Rank Displacement

25 equally-qualified (high-fit) candidates. Worst case: high-fit group writes with Llama-3.1-8B, moderate-fit group writes with a top proprietary model.



2 of 25

qualified candidates displaced purely by model mismatch.

+7.6% net displacement rate above the neutral (same-writer) baseline

Take-Away

An Inequity Built on Tool Access

FOR APPLICANTS

Tool choice carries real, unrecoverable consequences. Skipping AI is worse.

FOR EMPLOYERS

Fairer models exist and are identifiable. Screening-model choice is a design decision.